Influences on the mental health and well-being of retired professional athletes from high contact team sports: a mixed methods systematic review

Jordan D Vella, Elise R Facer-Childs, Jessica Ogden, Sarah Liddle, Alyssa Engelman, Ilan Volchek, Kylie King

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School of Psychological Sciences, Monash University Faculty of Medicine Nursing and Health Sciences, Clayton, Victoria, Australia

Correspondence to
Jordan D Vella;
jordan.vella@monash.edu

ERF-C and JO contributed equally.

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ABSTRACT

Objective To report the prevalence of mental health symptoms and influencing factors in retired professional high contact team sport (HCTS) athletes.

Design Mixed-methods systematic review.

Data sources PsycINFO, Embase, Medline, SPORTDiscus and Scopus were searched in July 2023 and March 2025.

Eligibility criteria for selecting studies Studies that investigated mental health and/or influencing factors within retired professional HCTS athletes were included. Studies that were non-peer-reviewed, could not obtain full text, used secondary data or focused on non-elite/individual/low-contact sports were excluded.

Results 85 studies were included in the final review, comprising 53 996 participants (females; n=247, 0.46%) from six sports (Australian Football League, Canadian Football League, football/soccer, ice hockey, National Football League and rugby). Prevalence ranges varied for each condition; smoking (0.9%–16%), depression (3%–49%), anxiety (4.3%–42%), cannabis use (5%–15.7%), adverse alcohol use (6.4%–68.8%), opioid use (7%–23.6%), stress (8.7%–26.9%), illicit drug use (10%–63.2%), anxiety/depression (10.2%–39%) and adverse nutritional behaviour (23.8%–64.5%). Of the studies including M and SD of validated scales, scores for depression, anxiety and sleep disturbance were equivalent to population norms, whereas mild or higher scores were reported for stress and adverse alcohol use. Concussion, pain, injury, neurological factors and declined physical function were shown to have a negative influence on mental health. Both negative and positive influences were observed for: athletic identity, psychosocial support, retirement autonomy, life events, osteoarthritis, retirement and cognitive function. 48% of studies had good methodological quality; however, most studies were cross-sectional, relied on self-report measures and lacked follow-up data and female athletes.

Conclusion Retired HCTS athletes experience high levels of psychological distress and adverse alcohol use, but depression, anxiety and sleep disturbance were similar to population norms. These results call for greater support measures for retired athletes.

PROSPERO registration number CRD42023449114

WHAT IS ALREADY KNOWN

⇒ Prevalence of mental health symptoms and disorders occurs in 16%–26% of retired elite athletes.
⇒ Stronger endorsement of identifying as an athlete has been associated with greater difficulties transitioning out of sport.
⇒ The prevalence of psychological distress in retired athletes is similar to the general population.

WHAT THIS STUDY ADDS

⇒ Prevalence of mental health symptoms ranged between 0.9% (smoking) and 68.8% (adverse alcohol use) in retired high contact team sport professional athletes.
⇒ Higher athletic identity during retirement can potentially help manage health conditions (eg, osteoarthritis).
⇒ This is the first study to report the negative influence of various neurological factors on mental health (eg, disrupted white matter integrity and deep white matter lesions were associated with cognitive dysfunction and depression).
⇒ Retired athletes with a history of concussion are at higher risk for adverse alcohol use and mood dysfunction compared with the general population with no history of concussion.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This systematic review highlights sports and athletic groups which are under researched, prompting further investigation into these areas.
⇒ The qualitative assessment of the included studies can help improve future research designs.
⇒ The recommendations from the authors can help influence sporting organisational policies in supporting retiring/retired athletes.

INTRODUCTION

Retiring from sports is an inevitable part of any athlete's career. While many athletes successfully transition out of their respective sports, a recent systematic review and meta-analysis found that mental health symptoms and disorders occurred in 16%–26% of elite athletes following retirement.[1] A variety of factors can contribute to career transitions, including concussion history (CH),[2 3] injury,[4 5] athletic identity (AI)[6]—how strongly one identifies as an athlete,[7] the achievement of sporting career goals, preretirement planning, involuntary retirement[8] and low levels of psychosocial support.[9]

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Athletes from high contact team sports (HCTS; sports where frequent, forceful, and intentional physical contact between players and equipment are permitted by, and integral to, the rules of the game) are vulnerable to post-retirement distress as these sports—rugby, football/soccer (hereafter referred to as football), ice hockey, lacrosse, American/Australian football—often involve higher rates of concussion[10–12] and injury,[5] which can lead to premature retirement and contribute to later life physical conditions, such as osteoarthritis (OA), chronic pain and psychological distress.[3 13] Additionally, athletes who retire from team-based sports can lose the camaraderie of a team environment and support of their teammates.[14]

Systematic reviews have been conducted on the transitional processes of retired athletes[6] and identified the prevalence of mental health symptoms and disorders[1 3] and their correlates[3 5 6 8 15 16]; however, these reviews often include a mixed sample of sports (eg, individual- and team-based/low contact to high contact/combat and weightlifting) at various competition levels (eg, Olympic, collegiate, elite, semiprofessional and professional), and often do not include qualitative research. This has led to uncertainty about the impact and experiences of retirement for specific athletic groups and how support methods can be improved.

The aim of this paper is to report the prevalence and experiences of mental health symptoms within retired professional athletes (ie, sports where financial compensation was provided for athletes' full-time performance, participation or competition) from HCTS and identify their influencing factors by synthesising relevant research from quantitative, qualitative and mixed-methods studies. To the best of our knowledge, this is the first mixed methods review on this topic. A mixed-methods design was chosen to broaden the literature in this area and answer the following questions: (1) What is the prevalence of mental health in former HCTS athletes? (2) What factors contribute to these rates? (3) What are the experiences of these athletes? and (4) How do quantitative and qualitative research compare in understanding the experiences of retired HCTS athletes?

METHODS

This systematic review was conducted following the Preferred Items for Systematic Reviews and Meta-analyses 2020 guidelines[17] and registered on the International Prospective Register of Systematic Reviews (PROSPERO; CRD42023449114). Abstract screening, full-text screening, quality assessment and data extraction were performed using the systematic reviewing management system, Covidence.[18]

Search strategy

A comprehensive search of relevant literature across five databases (PsycINFO, Embase, Medline, SPORTDiscus and Scopus) was conducted during July 2023, and again during March 2025, with no restrictions to year of publication or language (one study was translated via Google Translate but ultimately excluded). Individual text words were used with Booleans (OR/AND) and Medical Subject Headings (MeSH) terms. A combination of sport performance level (eg, 'Sport/ Athlete/ Professional Athlete/ Professional Sport/ (MeSH)' OR 'professional' OR 'elite' Adj2 'athlete*', 'player*', 'sport*') AND retirement status (eg, 'Retirement (MeSH)' OR 'retir*', OR 'career transition', OR 'former', OR 'ex', OR 'past') AND type of sport (eg, 'Football/ Rugby/ Soccer/ Contact Sport/ Collision Sport/ Team Sport (MeSH)' OR 'football', OR 'soccer', OR 'rugby', OR 'NFL', OR 'national football league', OR 'AFL', OR 'Australian football league', OR 'VFL', OR 'hockey', OR 'NHL', OR 'NAHL', OR 'contact', OR 'collision', OR 'team', OR 'lacrosse') were used. See online supplemental material 1 for specific search terms used across databases.

Study selection

Eligible studies that investigated mental health and/or influencing factors within retired professional HCTS athletes were included. Studies that were non-peer reviewed, used secondary data or the full-text could not be obtained, were excluded. Studies involving current or retired collegiate/ amateur athletes, or former athletes who competed in individual, low contact or combat sports as controls, were included.

Screening

All eligible studies were imported into the citation and reference management tool, EndNote V.20[19] and then uploaded into Covidence with duplicates being automatically/manually deleted during both processes. Two reviewers (JDV and JO) independently screened each study by title and abstract, and two pairs of reviewers (JDV/JO and JDV/ IV) independently performed a full-text screening against a predetermined protocol criteria.

Quality assessment

Appraisal of selected studies was performed independently by two reviewers (JDV and AE) using the National Institutes of Health (NIH) quality assessment tools[20] for quantitative studies and mixed-methods quantitative data and The Critical Appraisal Skills Programme (CASP) qualitative research assessment tool[21] for qualitative studies and mixed-methods qualitative data. The NIH assesses internal validity through a checklist of 'yes', 'no', 'cannot determine', 'not applicable' or 'unreported' against 12–14 prespecified questions, and reviewers determine if study quality is 'good', 'fair' or 'poor' against their own judgement. The CASP includes 10 questions against a checklist of 'yes', 'can't tell' or 'no' to help reviewers appraise study quality systematically. Any disagreements regarding quality appraisal were resolved through discussions between reviewers, with a third reviewer (JO) available if consensus could not be reached.

Data extraction

All data extraction was performed by two pairs of independent reviewers (JDV/JO or JDV/AE), with the following data being extracted to an Excel spreadsheet: study design, participants (number, biological sex, age, years retired, and sport played), measurements, mental health (depression, anxiety, depression/anxiety, quality of life, substance use, adverse nutritional behaviour, global mental health, distress, sleep disturbance, 'other') and influencing factors (AI, concussion, psychosocial support, education, career, neurological, pain, retirement decision, injury, life events, 'other'). For quantitative data, prevalence, mean (M) and standard deviation (SD), medians (Md) and mean differences (MD) were collected. For influencing factors, correlations, odd ratios (ORs), risk ratios (RRs) and prevalence ratios (PRs) were extracted. For qualitative data, themes/subthemes and direct quotes from each study were extracted by two independent

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Table 1 Quantitative and qualitative data synthesis

Theme	Subthemes	Findings
Athletic identity (AI) (n=8; 9%)	Negative influence (5/8)	Higher AI during retirement was associated with feelings of abandonment, rejection, insecurity, loss of future objective, distress and depression.
	Positive influence (3/8)	Higher AI can help manage conditions (eg, OA) through a non-defeatist attitude, while lower AI can promote better mental health by endorsing a new identity.
Concussion history (CH) (n=23; 27%) *7 found no relationship	Negative influence (17/23)	Undiagnosed concussions were the strongest predictor of opioid misuse in one study while a higher CH was related to depression, anxiety, substance use, psychosocial functioning and experiences of frustration and sleep disturbance.
Psychosocial support (n=12; 14%)	Negative influence (7/12)	Low social support from sporting supervisors, clubs and organisations influenced feelings and experiences of anger, hurt, disappointment, stress and burnout. Struggles with social isolation post-retirement were related to mild depression. Relationship issues, lack of social support and loss of community were associated with depression and grief.
	Positive influence (5/12)	Support from family, friends, counsellors, wives, team mentors, religious organisations and team staff supported mental and emotional health. Support from sport psychologists reduced stress. Support from sporting organisations influenced feelings of respect and appreciation when retiring.
Pain (n=11; 13%)	Negative influence (11/11)	Higher levels of pain intensity and interference were associated with feelings and experiences of frustration, bitterness, anger, irritability, sorrow, substance use, depression, anxiety, stress, anxiety/depression and lower cognitive QoL.
Retirement autonomy (n=14; 16%)	Negative influence (11/14)	Involuntary retirement was associated with feelings and experiences of anger, disappointment, sadness, lower self-esteem, stress, depression, anxiety, confusion, ashamedness and lower mental health.
	Positive influence (3/14)	Retiring voluntarily felt comforting, freeing, and like a natural process and was positively associated with better QoL and psychological flourishing.
Injury (n=9; 11%)	Negative influence (9/9)	Injuries were strongly related to experiences and feelings of discomfort, stress, loss of self, worries, concern for long-term health, sadness, grief, anger, frustration, depression, addiction, lower mental health, anxiety/depression, sleep disturbance and lower QoL.
Life events (LEs) (n=10; 12%)	Negative influence (8/10)	A higher number of LEs (including retirement, career transitions, and figuring out what to do post-retirement) were associated with feelings and experiences of anxiety, stress, depression, sleep disturbance and adverse nutritional behaviour.
	Positive influence (2/10)	Finding work and working more hours were negatively associated with distress and anxiety/depression, while positive transitions and career self-doubt were positively associated with psychological flourishing
Osteoarthritis (OA) (n=7; 8%)	Negative influence (5/7)	OA was associated with feelings and experiences of frustration, anger, bitterness, irritability, anxiety/depression, lower mental health, distress, sleep disturbance and substance use.
	Positive influence (2/7)	Not having OA resulted in better overall mental health.
Cognitive function (n=6; 7%)	Negative influence (5/6)	Reductions in cognitive function were associated with depression, anxiety, memory loss and ADD/ADHD.
	Positive influence (1/6)	Better cognitive function was associated with lower levels of depression and anxiety.
Physical function (PF) (n=6; 7%)	Negative influence (6/6)	PF decline was associated with strong feelings and experiences of loss, sadness, substance use and depression.
Retirement (n=10; 12%)	Negative influence (5/10)	Retirement can influence feelings and experiences of depression, anxiety, anger, loneliness/isolation and stress; as well as substance use due to the removal of routine testing. Removal of daily routine associated with negative feelings, and mental health positively associated with how planned and prepared athletes are for retirement.
	Positive influence (5/10)	Retirement removes sport-related stressors, including punitive discipline and sporting culture, bringing relief to retirees. Planning and preparing for retirement allowed athletes to retire with pride, sense of purpose, excitement and cultivate identities beyond sport.

Refer to online supplemental table 3 for a list of all measurement tools.
(n=x; y%) = number of studies contributing to conclusions of risk and protective factors; percentage of total number of studies. (x/n) = proportion of studies reporting construct as a risk or protective factor.
ADD/ADHD, attention deficit disorder/attention deficit hyperactivity disorder; QoL, quality of life.

reviewers (JDV/KK). A third reviewer (ERF-C) conducted a consensus review of all data extractions and resolved any discrepancies.

Data synthesis

JDV synthesised the extracted quantitative (online supplemental table 1) and qualitative data (online supplemental table 2) in consultation with ERF-C, SL and KK. The final themes are presented in table 1. Due to the heterogeneity of studies, a meta-analysis was not possible (see online supplemental table 3 for a list of materials). JDV then produced a narrative synthesis in consultation with ERF-C, SL and KK, synthesising all the data together noting points of consensus or divergence, as presented below with tables used to aid in data presentation. This data synthesis method followed the recommendations of Onwuegbuzie and Leech (2021) for qualitising data in mixed-methods research.[22]

Equality, diversity and inclusion statement

The author team consisted of two men and five women who were either students or early-career to mid-career researchers. This research examines the mental health of retired athletes from diverse age groups and ethnic backgrounds, who have been employed in various sports and different countries. This under-researched population is known to experience a range of mental health symptoms and disorders. Of the studies that reported ethnodemographic data, white/Caucasian participants made up the majority (62%), while black participants made up 34%, and other minority/non-white groups made up 4%.

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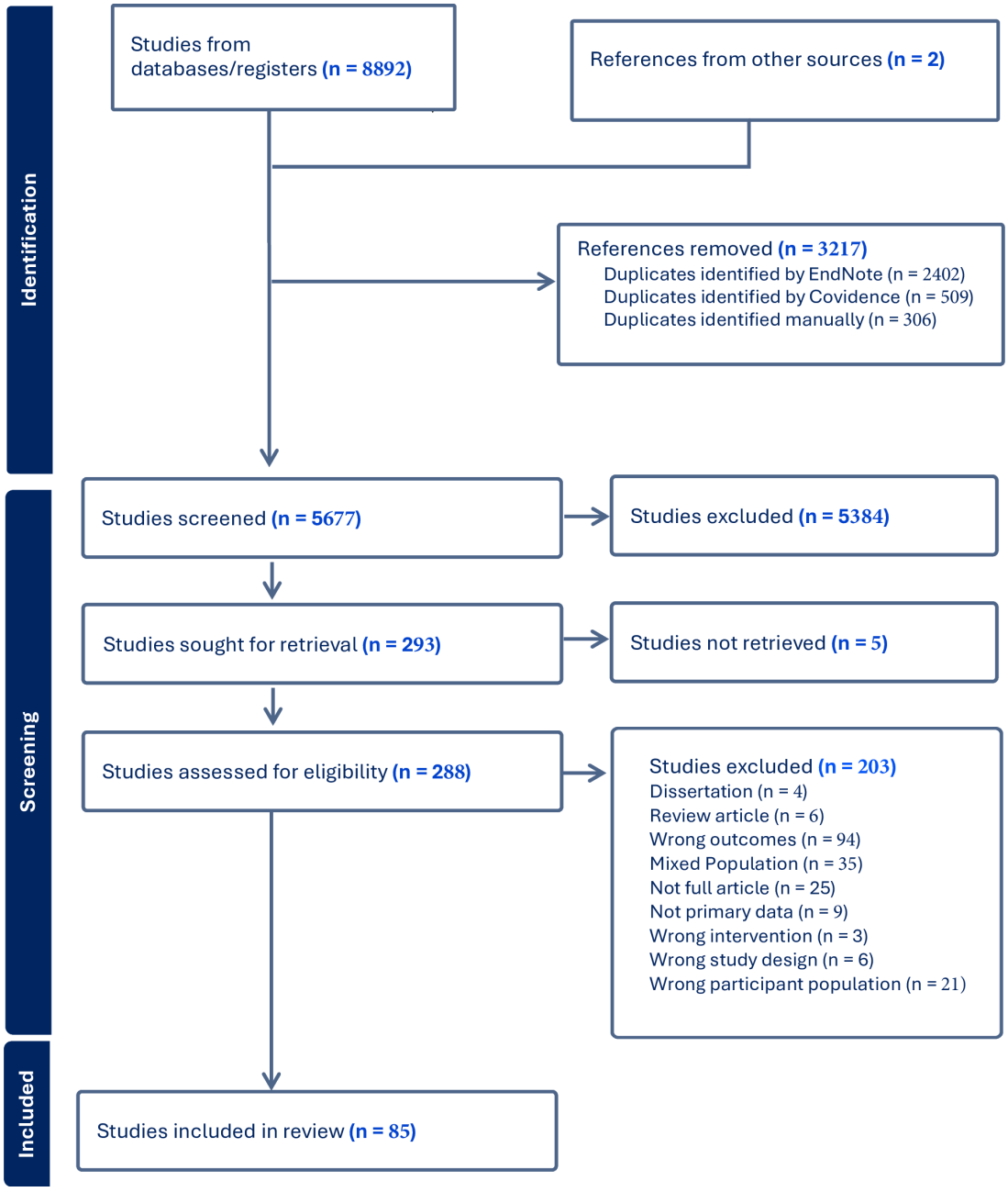

Figure 1 Study selection flowchart.

RESULTS

Study selection

After 3217 duplicates were removed, the initial search yielded a total of 5678 studies. Then, 5384 abstracts were excluded after screening titles and abstracts against the eligibility criteria, leaving 294 full-text articles to be reviewed for inclusion; 209 studies were excluded for various reasons as outlined in figure 1. Finally, 85 studies were included in this review. The Results and Discussion sections refer to findings from self-reported mental health symptoms unless specified otherwise.

Study characteristics

Studies were conducted in North America (52%; n=44), Europe (24%; n=20), Australia (9%; n=8), South America (1%; n=1) and the Middle East (1%; n=1), and 13% involved a mix of intercontinental participants (n=11). Quantitative studies constituted 86% of studies (case–control, n=3; cohort, n=9; cross-sectional, n=52; cross-sectional case–control, n=1; cross-sectional cohort, n=2; randomised control trial (RCT), n=1; longitudinal, n=5), while 12% were qualitative (n=10), and 2% were mixed methods (n=2). Twenty-eight studies included comparative analyses (reported below), including eight studies which compared retired and current HCTS athletes. However, as the focus of this paper is on retired athletes, no data relating to these comparisons have been included. A variety of sports comprised the sample, including the National Football League (NFL; n=36), football (n=26), rugby (league and union; n=15), ice hockey (n=9), Canadian Football League (CFL; n=3), Australian Football League (AFL; n=3), and one study also included former cricket and Gaelic sports players. Data were collected from 53 996 participants (age range=19–92 years; females, n=247, 0.46%) who had been retired for an average range of 2.25–29.2 years.

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Of the studies, 95% included only male participants (n=81), 2% included both males and females (n=2) and 2% included females only (n=2). Five studies reported 'non-white', 'white' and 'person of colour' categories. Twenty-nine studies provided adequate ethnodemographic data grouped by 'white/Caucasian', 'black/African American' and 'other' (mixed, American Indian, Alaskan Native, Asian, Hawaiian native or other pacific islanders and Hispanic). The total number of participants included was 40 597. Of the studies that reported ethnodemographic data, participants identifying as white/Caucasian comprised 61.56% (n=24 992), black/African American 34.30% (n=13 923), other 2.54% (n=1031) and 'non-white' or 'person of colour' 1.60% (n=651).

Risk of bias in studies

Forty-one studies had good methodological quality (48%), forty-one had moderate methodological quality (48%) and three had poor methodological quality (4%). Of the 73 quantitative studies, 32 were of good quality (44%), 39 were of moderate quality (53%) and 2 were of poor quality (3%). The main issues were lack of sample size, power descriptions, follow-ups and blinding. Of the 10 qualitative studies, 7 had good methodological quality (70%), 2 had moderate quality (20%) and 1 was of poor quality (10%). The main issue was a lack of consideration for the relationship between researcher and participants. Both mixed-methods studies had good methodological quality (100%). For full-quality appraisal, see online supplemental tables 4-7. Additional analysis was carried out on only high-quality studies finding that prevalence rates remain similar (*see* online supplemental table 8).

Prevalence and incidence rates

Prevalence ranged from 0.9% for smoking to 68.8% for adverse alcohol use (see table 2). Studies that included self-reported lifetime clinical diagnoses/history, where the condition onset was not specified, were not included as it is unclear if these occurred postretirement. Incidence rates of self-reported mental health symptoms were reported in three studies, with one study reporting 12-month incident rates of anxiety/depression (29%), adverse alcohol use (15%), stress (11%) and sleep disturbances (28%) in former football players,[23] another study reporting 6-month incidence rates of anxiety/depression (8.3%), adverse alcohol use (16.7%), adverse nutritional behaviour (8.3%), stress (8.3%) and sleep disturbances (25%) in former ice hockey players,[24] and one longitudinal study found a significant increase of depression diagnoses (11.2%) between 2001 and 2019.[25]

Self-reported mental health symptoms

Twenty-five studies (29%) reported M and SD depression scores, with 15 studies finding minimal levels of depression,[25–39] nine studies finding mild–moderate levels of depression[40–48] and one mixed-methods study reporting a range of mild–extremely severe depression.[49] Nine studies reported M and SD anxiety scores, eight studies found minimal levels of anxiety,[28–30 32 34 37 40 46] two studies found mild anxiety within former rugby players[41 45] and one study found moderate anxiety in former Iranian footballers.[48] Six studies reported M and SD stress scores, three studies found moderate levels of stress in former rugby players,[40 41 45] one study found high levels of stress in former Iranian footballers[48] and two studies found low-average stress levels in former rugby players.[28 34] All six studies reporting M and SD alcohol consumption scores included former rugby players, with four studies reporting hazardous consumption levels[27 34 40 45] and two studies reporting low-risk consumption.[30 41] Only four studies reported M and SD sleep scores, finding normal levels in former rugby[40 41] and NFL players.[28 50] Twelve studies reported M and SD mental health component scores, six studies found average scores in former NFL and football players,[48 51–55] two studies found below-average scores[56 57] and four studies reported above-average Md scores.[58–61]

Comparative studies

Retired athletes from HCTS versus retired athletes from non-contact (NC) sports. Five studies compared retired HCTS athletes to retired NC sport controls (football, n=2; rugby, n=1; NFL/ice hockey, n=2). Depression was reported to be significantly higher in retired football, rugby and NFL/ice hockey athletes[27 33 35 46 62]; however, in three studies, both groups scored within the 'normal/minimal' range.[33 35 46] Two studies found no significant differences in anxiety between former football[33] and rugby players[27] and controls. In contrast, one study reported significantly higher anxiety in former NFL and ice hockey players (but both still scored in the minimal range).[46] One study found lower anxiety/depression in retired handballers compared with retired footballers, while also reporting lower stress and sleep disturbance, and higher levels of alcohol consumption, in former footballers.[62] One study found no significant difference in alcohol consumption between retired rugby players and NC

Table 2 Prevalence

	NFL	Rugby (league/union)	Football (soccer)	Ice hockey	Total
Depression	3%–31.6%(14)	4%–49%(4)	4.3%–15.6%(4)	31.3%(1)	3%–49%(23)
Anxiety	4.3%–34.8%(9)	14%–42%(3)	11.3%–12%(2)	–	4.3%–42%(14)
Anxiety/depression	12%–30%(2)	28.4%–28.8%(2)	10.2%–39%(5)	19.4%(1)	10.2%–39%(10)
Adverse alcohol use	6.4%–51.6%(7)	23.8%–59%(6)	8.4%–68.8%(4)	28.7%(1)	6.4%–68.8%(18)
Adverse nutritional behaviour	–	61.9%(1)	40%–64.5%(3)	23.8%(1)	23.8%–64.5%(5)
Stress	–	10%–26.9%(4)	8.7%–26.3%(4)	11.7%(1)	8.7%–26.9%(9)
Sleep disturbance	–	17%–35%(3)	11%–33.04%(4)	16.7%(1)	11%–35%(8)
Opioid use	7%–23.6%(3)	–	–	–	7%–23.6%(3)
Cannabis use	15.7%(1)	5%–5.3%(2)	–	–	5%–15.7%(3)
Illicit drug use	63.2%(1)	12.8%(2)	10%(1)	–	10%–63.2%(4)
Smoking* (current)	0.9%–16%(4)	4%–15%(2)	11.4%–12%(2)	–	0.9%–16%(8)

Contributing studies = [24 26–30 40–44 50 51 58 62 65 68 70–73 75 79 85–87 91 92 94–96 99–104 108 109]. (n) = number of studies contributing to prevalence rates presented. *Refer to* online supplemental table 3 *for a list of all measurement tools.*
*Reported as a current tobacco smoker.
NFL, National Football League.

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sport controls but reported significantly higher sleep disturbance in retired rugby players.[27] Global mental health was similar between retired female footballers and controls.[33]

Retired athletes from HCTS with a history of concussion versus general population (GP) with no history of concussion. Seven studies compared retired athletes to healthy controls (ie, no history of concussion; NFL, n=2; rugby, n=3; CFL, n=1; ice hockey, n=1). Three studies found no significant differences in depression and anxiety between former rugby[34 45] and CFL athletes[32] and controls, while one study found significantly higher depression in former rugby players; however, it is important to note that both groups scored within the 'normal/minimal range'.[31] Two studies found higher but non-significant differences in stress and significantly higher levels of alcohol consumption, between former rugby players and controls.[34 45] Compared with controls, one study found significantly greater mood/behaviour dysfunction in former NFL players,[63] while one study reported significantly higher psychosocial endorsement by retired ice hockey players.[64] One study found significantly higher self-reported diagnosed sleep apnoea and daytime sleepiness in retired NFL players compared with controls; however, both scores fell within the normal range.[50]

Retired athletes from HCTS versus GP. Eight studies compared retired HCTS athletes to GP controls (NFL, n=3; football, n=2; rugby, n=2; football/rugby, n=1). Two studies found no significant differences in depression and anxiety between former football[51] and rugby players[30] and controls, while two studies found former NFL players reported significantly higher levels of depression[36 44]; however, in the former study, both groups scored within the 'normal/minimal' range.[36] One study found that former NFL players reported significantly lower anxiety than controls,[65] while one study found no significant differences in illicit drug use between former NFL players and controls.[44] One study found that former NFL players consumed significantly higher levels of alcohol compared with controls[44] and one study reported no significant difference between former rugby players and controls (both low risk).[30] One study found former footballers reported significantly higher prevalences of both sleep disturbances and optimal sleep (7–8 hours).[51] Two studies found similar mental health scores between former football and rugby players compared with controls[53 59] while two studies found significantly lower mental health scores in retired football players (although this was still similar to US population norm)[51] and rugby players[61] compared with controls.

Influencing factors

AI negatively impacted mental health in five studies, including three qualitative studies,[14 66 67] one mixed-methods study[49] and one quantitative study (OR=1.28) indicating that one unit increase in AI was associated with a 1.28 unit increase in depression.[68] In contrast, two qualitative studies[14 69] demonstrated how lower *and* higher levels of AI can be beneficial to mental health, while one quantitative study showed a significant negative correlation between AI and psychological flourishing.[60]

Seventeen studies reported that CH negatively impacted mental health. These included one qualitative study[67] and 16 quantitative studies.[25–28 31 36–38 41 56 65 70–74] In one study, self-reported undiagnosed concussions were the strongest predictor of opioid misuse (aOR=4.25) in multivariate analyses.[70] However, findings should be replicated to strengthen this conclusion as CH was based on participants' opinion, potentially limiting the internal validity of results. One study reported a higher PR for developing depression with three or more concussions (PR=2.58) or 1–2 concussions (PR=1.38).[56] One longitudinal study reported a linear dose relationship between CH and the nine-year risk of depression diagnosis, ranging from 3% with zero concussions to 26.8% with a history of 10 or more concussions[74]; however, another longitudinal study found that while former NFL players with a higher CH were more likely to report depressive symptoms, CH alone did not significantly predict the trajectory of symptom severity over 19 years.[25] It is important to note that most of these studies were cross-sectional (75%), relied strongly on retrospective self-reported data, and only 62.5% reported that a clear definition of concussion was provided to participants. Increased risk of inaccuracy and recall bias should be taken into consideration when interpreting the findings. Contrastingly, seven studies found no relationship between CH and mental health[30 32–34 40 75 76] attributing this to mild traumatic brain injuries, high resilience, better mental health scores, low pain interference and higher physical functioning. Interestingly, one study found that increased heading exposure (football technique) was associated with worse depression, anxiety and mental health in former female footballers but not CH.[33]

Seven studies found lower levels of psychosocial support negatively impacted mental health, including two qualitative studies,[77 78] one mixed-methods study[49] and four quantitative studies.[42 43 56 79] Two studies reported ORs, showing an association between low supervisor support and distress (OR=1.0) and burnout (OR=1.1)[79] and relationship/marital problems (OR=4.6), lack of friendships (OR=5.6) and loss of community (OR=5.3) with depression.[42] In juxtaposition, four qualitative studies found increased social support positively impacted mental health[14 66 77 80] and one mixed-methods study found being a parent was an important factor in preventing a suicide attempt in a former NFL player.[81]

Nine studies found various neurological factors negatively impacted mental health, including eight quantitative studies[26 39 45 57 64 82–84] and one mixed-methods study.[81] Neurological associations included increased grey matter myo-inositol (mL), glutamate (Glu), N-acetyl aspartate (NAA), creatine and glutamate/glutamine with depression, stress and anxiety; and higher levels of white matter glutamate with alcohol consumption.[45] Pituitary hormone dysfunction and poor quality of life (QoL)[57]; disrupted white matter integrity and deep white matter lesions with cognitive dysfunction and depression[82]; worse mood/behaviour with increased right thalamic volume[83] and higher concentrations of Glu, glutathione, mL and NAA in the anterior cingulate gyrus.[84] The presence of the apolipoprotein allele and worse psychiatric complaints[64]; total fractional anisotropy (FA) injury burden of the brain and worse emotional well-being[39]; white matter tracts FA and depression (particularly the forceps minor)[26]; and genetic dispositions.[81]

Pain negatively impacted mental health in one qualitative study[69] and nine quantitative studies.[41 42 47 68 70 72 73 85–87] Six studies reported ORs with one study finding higher levels of pain associated with opioid use (OR=4.5).[70] Three studies reported the relationship between pain and depression (OR=1.38),[68] (OR=3.5)[42] and (OR=0.46) respectively,[47] and one study reported adjusted ORs for pain intensity and anxiety/depression (aOR=0.27), and pain interference and anxiety/depression (aOR=2.29) and anxiety alone (aOR=1.07).[85] Additionally, one study reported ORs for pain severity (OR=0.31) and pain interference (OR=0.47) with moderate–severe depression.[86] Moreover, one study reported pain intensity to be a negative influence for opioid use only (aOR=1.20) and co-use of opioids and sedatives (aOR=1.58).[72]

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Eleven studies found involuntary retirement negatively impacted mental health, including six qualitative studies,[14 67 78 88–90] four quantitative studies[37 55 91 92] and one mixed-methods study.[49] One study included ORs for distress (OR=2.10) and comorbidity of common mental disorders (OR=1.89).[91] Contrastingly, one qualitative study showed mental health benefits,[90] and two quantitative studies showed better QoL[71] and psychological flourishing[60] when voluntarily retiring.

A history of physical injury during playing careers, and/or its influence as a reason for retirement or long-term pain, unanimously impacted mental health negatively. This was supported by two qualitative studies[77 89] and seven quantitative studies.[27 47 52 62 68 70 93] Three studies provided ORs, with one study finding an association between 3+ injuries and opioid use (OR=6.02),[70] one study reporting an OR of 3.44 for depression if sustaining a career-ending injury,[68] and one study found associations between anxiety/depression (OR=1.1) and sleep disturbances (OR=1.2) if reporting higher sustained severe injuries.[62]

Increased major life events (LEs) (eg, grievance, change of financial status, change in work hours, birth of a child) negatively impacted mental health. This was supported by two qualitative studies,[66 88] one mixed-methods study[81] and five quantitative studies.[23 42 62 79 94] One study reported RRs for sleep disturbance when experiencing three or more LEs (RR=4.52) or 1–2 LEs (RR=1.73),[23] and one study reported ORs for a higher number of LEs and distress (OR=1.2), sleep disturbance (OR=1.6), adverse nutritional behaviour (OR=1.8) and anxiety/depression (OR=1.6).[94] Two studies reported ORs for a higher number of LEs in the previous 12 months and distress (OR=2.0), sleep disturbance (OR=1.5) and anxiety/depression (OR=1.4).[62 79] One study reported the OR for depression was higher if experiencing postretirement transition difficulties (OR=5.4) or having difficulties finding work or meaningful employment (both OR=4.9).[42] In contrast, one quantitative study reported lower ORs for distress (OR=0.6) and anxiety/depression (OR=0.6 and OR=1.0) when finding employment post-sport and working longer hours,[95] while another quantitative study found psychological flourishing to be negatively correlated with career self-doubt and positively correlated with a positive retirement transition.[60]

One qualitative study[69] and four quantitative studies[53 54 96 97] found the presence of OA was associated with worse mental health. In contrast, two quantitative studies reported no OA resulted in better overall mental health.[53 54]

Five studies found a decrease in physical function (PF) to be associated with worse mental health including one qualitative study,[69] and five quantitative studies.[25 47 70 73 74] One study reported an association between physical impairment and opioid misuse (OR=3.49) and found that misusers were more likely to need a cane, wheelchair or walker (OR=7.59).[70] One longitudinal study found the nine-year risk of depression increased by 40% for every 10-point decrease in PF,[74] and one longitudinal study reported that the rate of PF decline, above CH, was the strongest predictor of depression symptom endorsement over a 19-year period.[25] Higher PF was also found to be negatively correlated with depression and anxiety.[73]

Five studies demonstrated that the initial retirement period can negatively impact mental health, including four qualitative studies[14 80 88 89] and one quantitative study.[37] In contrast, three qualitative studies,[77 80 88] one quantitative study[92] and one mixed-methods study[81] highlighted that retirement can be beneficial to mental health. One study reported the nine-year risk of depression diagnosis in former footballers decreased 17% for every 10-year increase in retirement years,[74] while another study showed a positive correlation between age of retirement and mental health.[98] The main positive infleunces for some athletes' initial retirement included: the removal of sport-related stressors, having planned and prepared for retirement, embracing a more holistic identity (not just athletic), being able to talk to older players with similar experiences, increased psychosocial support and better coping strategies. The main negative infleunces for some athletes, included: involuntary retirement, no plan for postretirement, significant changes and disruption to AI, change in social status, loss of social connection with teammates, sport culture promoting stoicism (ie, emotional avoidance) and reduced help seeking, impacts of injury, insufficient support from clubs, removal of daily routine and poor coping mechanisms (eg, alcohol use).

Five quantitative studies found that reduced cognitive function impacted mental health negatively.[65 71 76 86 99] One study reported an association between daily problems in cognitive function and a diagnosis of depression (aOR=6.17), anxiety (aOR=5.06) or attention deficit disorder/attention deficit hyperactivity disorder (aOR=4.73),[99] while in another study using multivariable models, anxiety was the only mental health variable to remain having a significant association with subjective cognitive difficulties.[76] In contrast, one study reported better cognitive function to positively impact mental health.[73]

Other negative influences for mental health include a history of opioid misuse with current opioid misuse[70 75]; the intercorrelation between depression, anxiety[73] and stress[41] symptoms; increased sleepiness and depression[50]; chronic traumatic encephalopathy and depression/ sleep apnoea[100] and suicidality[101]; normative male alexithymia (limited ability to articulate and express emotions developed from masculine traditions to suppress emotions) and depression[43]; higher number of surgeries and anxiety/depression[62] and adverse alcohol use[24]; lower olfactory function with worse behavioural/mood symptoms[44]; stoicism and masculine culture associated with alcohol use[77] and distress[14]; career dissatisfaction increasing odds of distress, smoking, sleep disturbance and adverse nutritional behaviour[24 94]; career duration with elevated odds of mental health problems (specifically 4–8 years)[102]; satisfaction with current income and career[103]; race (ie, being a minority players)[73 85]; and a higher number of adverse childhood experiences with depression and cognition QoL.[86] One study also reported current depression and anxiety symptoms to be significantly greater in participants who were 'exposed' to NFL younger than 12 years of age; however, this did not remain statistically significant in any linear regression models.[29]

Other positive influences on mental health include having higher levels of resilience[40 41] and education,[47 49] shorter or longer career durations (specifically less than 4 years and greater than 8 years) with less mental health problems[101 102] and meeting physical activity recommendations.[101] In the only RCT included in this review, an 8-week mindfulness-based stress reduction course significantly reduced symptoms of stress, anxiety and depression in the intervention group compared with the control condition, with improvements lasting 12-weeks postintervention.[48]

DISCUSSION

This paper reviewed the existing evidence in quantitative, qualitative and mixed-methods research regarding the prevalence and experience of poor mental health symptoms within retired professional HCTS athletes and their influencing factors. An extensive search strategy and selection process yielded 85

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original studies (48% with good methodological quality) that presented data on depression, anxiety, stress, sleep disturbance, substance use, adverse nutritional behaviour, mental health, QoL and smoking. Prevalence ranged from 0.9% for smoking to 68.8% for adverse alcohol use. The main factors influencing mental health included AI, concussion, psychosocial support, pain, retirement autonomy, neurological factors, injury, LEs, OA, retirement, and cognitive and PF. Average (M/SD) scores typically fell within the 'normal/minimal' levels of depression, anxiety and sleep disturbance, while mild or higher levels of stress and adverse alcohol use were observed.

Mental health symptoms were similar to retired NC athletes and healthy/GP controls (except depression and sleep disturbance which were higher in retired HCTS athletes when compared with retired NC athletes; and alcohol use, which was higher in retired HCTS athletes in comparison to healthy controls). Higher sleep disturbances can negatively affect mood and vice versa, and as both groups (HCTS and NC) are retired, similar experiences of identity conflicts, career transitions and loss of support would be expected. Comparisons between HCTS athletes and healthy/GP controls were also based on limited evidence (n=9), therefore, fewer studies highlighting the association between CH and mental health were included—and as most former athletes reported experiencing 'normal/minimal' levels of anxiety and depression—this may reflect the current findings.

Our findings of high prevalence rates of adverse alcohol use (eg, 68.8%) among retired HCTS athletes[104] could be attributed to the removal of routine testing following retirement[89] and the common use of alcohol as a coping mechanism in men.[105] This could be further supported by one study in this review, which examined incidence rates of alcohol use over 6 months and found these to be higher among retired athletes compared with current athletes.[24] These findings support previous study findings,[106] which reported similar rates between retired elite athletes and the GP, but contrast an earlier systematic review and meta-analysis, which found higher rates in current athletes.[1] The prevalence of smoking found in our data (0.9%) is much lower than general global prevalence rates reported (~17%).[107] This could suggest that participation in sport at any stage of life may act as a protective factor for health. Although this was not investigated in this review, future research could consider how participation in sport or physical activity could be protective against both mental health issues and adverse health behaviours.

The initial transition period out of a professional sporting career can create mixed emotions for athletes. To better prepare for a more positive transition following retirement, it is important that athletes sufficiently plan and prepare for retirement during their career, establish a daily routine following retirement, and develop a self-identity and social relations outside of sport. Sporting organisations can also support athletes by promoting and fostering alumni events and connections, and educate athletes on effective coping strategies (eg, cognitive restructuring, stress management and arousal regulation) to better manage stress. Both athletes and sporting organisations should also prioritise the continued educational pursuits of athletes during their career.

To the best of our knowledge, this is the first review to address the influences of neurological factors and the potential benefits of higher AI during retirement in helping to manage health conditions (though this needs further literary support). A significant strength of this study is the combination of quantitative and qualitative data. As far as we are aware, this is the first systematic review on mental health in retired athletes that includes qualitative studies that have considered athlete experiences. Past studies often report separately on quantitative and qualitative measures, but there is a need to present integrated findings that may help uncover additional information and insights of retired athlete experiences and mental health. For example, the influence of AI, retirement autonomy, and the initial retirement experience is predominantly supported by qualitative research. Without these studies, rich data pertaining to retired athletes' experiences would not be available.

The main limitations of the included studies are that most are cross-sectional in design and use self-report measures, which prevents the determination of causal relationships, increases the risk of bias, and only provides current symptomology. The plethora of scales and measurements also make comparisons between studies difficult, and the limited number of longitudinal studies reduces the certainty of associations between mental health and influencing factors over time. Additionally, as most studies included males only, the general retirement experiences of females are unclear. Our review found that females accounted for 0.46% of all participants, highlighting the underrepresentation of females in this field of research. Finally, basing conclusions on self-reported concussion and injury history, including their frequency of severity, further increases the risk of bias and needs to be acknowledged.

Future research should investigate causal relationships between influencing factors and mental health outcomes, and include more female participants (particularly retired female athletes), comparison groups and more qualitative methods to gain an in-depth understanding of the retirement experiences of professional HCTS athletes. There is also a potential need for more tailored measurements to be used for this population and a higher focus on highly impactful factors (eg, AI) with currently low literature support. In addition, most studies on athlete and retired athlete populations are applied research studies that face challenges with achieving the same rigour as some other clinical or lab-based research, as highlighted by our quality assessment. Future research should prioritise more robust methodology by addressing current issues with methodological quality and include the use of medical records and longitudinal studies for more accurate results.

We recommend that existing support guidelines are modified to include regular check-ins with athletes postretirement, as many athletes (especially those who retired involuntarily) in the studies reviewed, indicated a perceived lack of care from sporting organisations during their career transition, creating negative feelings and increasing their stress.[77 78 80] Moreover, a lack of support from sporting organisations can cause former athletes to disengage from support programmes putting them at higher risk for mental ill-health.[80] Sporting organisations should also ensure that athletes are adequately planning and preparing for their retirement as this ensures a more successful retirement transition.[80 81]

CONCLUSION

This review found a prevalence range of 0.9% for smoking to 68.8% for adverse alcohol use. Self-reported symptoms for depression, anxiety and sleep disturbance were in normal ranges, while higher levels of stress and alcohol use were observed. The findings of quantitative and qualitative studies generally supported each other; however, the quantitative literature found a greater range in mental health symptom severity, while the qualitative literature provided more depth regarding the impact of influencing factors. Due to the lower number of high-quality studies, and the primary reliance on self-report measures, it is

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recommended that findings are interpreted with caution. We recommend sporting organisations establish a duty of care for improved support during retirement, promote alumni social and support networks, and educate athletes on healthy coping strategies. We also recommend active athletes develop an identity and social relationships outside of sport, sufficiently plan and prepare for their retirement, and establish a daily routine following their retirement. These findings highlight some important targets for how athletes and organisations can create a positive transition to retirement.

Social media Jordan D Vella, LinkedIn @jordan-vella-339510288

Contributors JDV: conceptualisation, methodology, formal analysis, data curation, writing—original draft, guarantor. ERF-C: conceptualisation, formal analysis, data curation, writing—review and editing, supervision. JO: formal analysis, data curation. SL: conceptualisation, writing—review and editing, supervision. AE: formal analysis, data curation. IV: data curation. KK: conceptualisation, writing—review and editing, supervision.

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ORCID iDs
Jordan D Vella http://orcid.org/0009-0004-3299-2523
Elise R Facer-Childs http://orcid.org/0000-0003-4549-9266
Sarah Liddle http://orcid.org/0000-0001-6238-8635

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